

GARISSA, Kenya

WMO: 637230

Lat: 0.464S

Lon: 39.648E

Elev: 486

StdP: 14.44

Time Zone: 3.00 (E03)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	70.8	71.7	54.8	65.1	89.8	56.9	70.2	89.4	33.3	83.7	30.2	84.5	4.3	180	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	19.8	99.4	74.7	98.4	74.7	97.2	74.6	79.1	90.2	78.3	89.3	77.8	88.7	6.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.8	142.0	82.1	75.9	137.9	81.5	75.4	135.4	81.1	43.0	90.1	42.2	89.8	41.6	89.3	93.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.8	22.3	19.4	DB	59.9	102.3	6.7	2.0	55.0	103.7	51.1	104.9	47.4	106.0	42.5	107.4
			WB	58.8	82.0	4.8	1.5	55.3	83.1	52.5	84.0	49.8	84.9	46.2	86.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	84.3	85.3	86.7	88.0	86.9	84.6	82.2	80.9	81.3	82.8	84.8	84.2	84.2
	DBStd	2.82	1.64	1.67	1.68	2.06	1.97	1.73	1.65	1.58	1.62	2.07	2.25	1.97
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12523	1094	1027	1178	1108	1073	967	959	969	984	1078	1026	1060
	CDD65	7048	629	607	713	658	608	517	494	504	534	613	576	595
	CDH74	82968	7940	8135	9769	8483	7301	5380	4657	4857	5744	7268	6506	6928
	CDH80	41006	4164	4544	5535	4414	3497	2311	1915	2085	2749	3620	2943	3229
(14) Wind	WSAvg	8.3	4.5	5.7	6.6	8.4	10.1	11.6	11.7	11.6	11.0	8.9	5.0	4.0
(15) Precipitation	PrecAvg	14.1	0.6	0.1	1.5	2.7	0.6	0.2	0.2	0.3	0.2	1.3	4.2	2.2
	PrecMax	36.4	9.7	0.6	6.8	6.9	3.0	1.3	0.7	3.0	2.0	6.6	13.8	6.9
	PrecMin	5.1	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.1
	PrecStd	6.8	1.9	0.2	1.6	1.7	0.8	0.3	0.2	0.6	0.4	1.7	3.0	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.9	100.7	101.0	99.6	97.0	93.6	92.5	93.2	95.3	97.2	97.0	97.0
		MCWB	72.8	73.8	75.3	76.1	75.5	72.1	71.0	70.8	72.3	74.2	75.5	73.6
	2%	DB	97.4	99.0	99.8	98.2	95.3	92.3	91.3	91.6	93.8	96.1	95.2	95.3
		MCWB	73.3	73.9	75.1	75.9	75.2	72.1	70.9	70.4	71.6	73.9	75.3	74.2
	5%	DB	96.2	97.9	98.6	96.8	94.0	91.2	89.9	90.3	92.7	94.9	93.7	93.8
		MCWB	73.7	73.9	75.2	75.8	74.8	72.1	70.6	70.2	71.3	73.7	75.4	74.7
	10%	DB	94.7	96.5	97.1	95.2	92.6	89.8	88.6	89.2	91.3	93.4	92.0	92.2
		MCWB	74.1	73.9	75.2	75.9	74.5	71.8	70.4	70.1	71.2	73.4	75.5	75.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.8	79.0	79.7	80.3	78.6	76.2	73.8	74.6	75.6	77.7	79.2	79.5
		MCDB	91.6	93.7	93.6	90.5	92.0	86.8	87.3	85.3	88.9	89.7	89.0	88.4
	2%	WB	77.4	77.1	78.3	79.0	77.4	74.7	72.9	72.9	74.2	76.4	78.2	78.3
		MCDB	90.7	92.8	90.9	89.5	89.4	86.3	86.0	84.9	87.4	88.3	88.0	87.4
	5%	WB	76.5	76.1	77.6	78.2	76.5	73.9	72.2	72.0	73.2	75.7	77.4	77.6
		MCDB	89.9	91.9	89.9	88.5	88.0	85.4	84.6	84.0	86.0	87.1	87.0	86.8
	10%	WB	75.8	75.4	77.0	77.5	75.8	73.1	71.5	71.2	72.4	75.2	76.8	77.0
		MCDB	88.9	91.1	89.2	87.5	87.2	84.6	83.2	82.9	85.1	86.3	86.1	86.2
(35) Mean Daily Temperature Range	5% DB	MDBR	20.0	21.1	19.8	17.8	17.3	16.9	17.1	17.4	18.8	18.5	16.2	16.8
		MCDBR	22.1	22.8	21.0	19.2	18.7	18.2	18.6	18.6	20.0	19.8	18.3	19.0
		MCWBR	5.5	5.3	4.2	3.9	3.9	4.2	4.3	3.7	4.1	4.1	3.8	4.5
	5% WB	MCDBR	18.9	20.8	19.1	17.2	17.2	16.5	16.9	17.0	18.5	17.9	15.9	16.0
		MCWBR	5.4	5.4	4.5	4.5	4.5	4.8	4.7	4.5	4.8	4.6	4.5	4.8
(40) Clear-Sky Solar Irradiance	taub		0.463	0.479	0.466	0.417	0.409	0.471	0.469	0.430	0.409	0.410	0.437	0.465
	taud		2.211	2.152	2.218	2.396	2.419	2.182	2.176	2.301	2.350	2.391	2.336	2.233
	Ebn at Noon		276	273	274	281	276	254	257	274	286	289	282	274
	Edh at Noon		48	51	48	39	36	45	46	42	41	40	42	46
(44) All-Sky Solar Radiation	RadAvg		1865	1967	1990	1886	1781	1627	1639	1732	1879	1931	1797	1787
	RadStd		92	88	63	62	54	66	57	54	69	60	91	75

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page